Class 8: Drude Model: Thermal Conductivity

There are several instances in Engineering and Technology, where heat transfer is a very important aspect for the successful functioning of a product.
There are three modes for the transfer of heat

1) Conduction
2) Convection
3) Radiation

Let us examine these a little closer.

Conduction: This is the process of transfer of heat energy through lattice vibrations, from atom to atom, and through electrons in the solid. During this process atoms are largely fixed to their respective lattice locations, and there is virtually no large scale movement of atoms. We typically consider conduction as the primary mode of transfer of heat within a solid.

Convection: This is the process of heat transfer that we associate with liquids and gases. This process of heat transfer involves significant atomic movement. Consider, for example, the beaker shown in Figure 8.1 below.

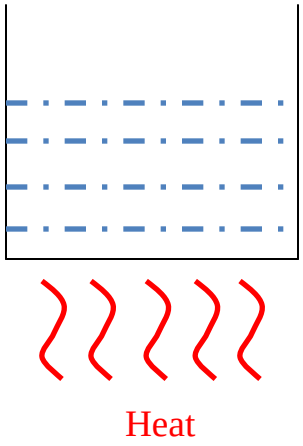


Figure 8.1: Schematic of a beaker containing a liquid, being heated slowly from the bottom

As heat is introduced into the beaker from the bottom, the temperature of the liquid close to the bottom increases. In general, as the temperature increases, the density of the liquid decreases. Therefore the density of the liquid closer to the bottom of the beaker decreases relative to the density of the same liquid higher up in the same beaker. This variation in density causes the liquid in the bottom to rise up and the liquid at the top to sink. The liquid and the atoms of the liquid therefore get significantly mixed up in this process, and heat gets transferred from the bottom to the top. This process of mixing up of the liquid, is called convection. If it occurs in the manner described above, it is called natural

convection. We can also speed up the process by adding stirrers or blowers to mix up the liquid faster – and hence transfer the heat faster. Such a process is called forced convection.

A day-to-day example where we see forced convection is in the functioning of air conditioners. In principle we can have the air conditioner provide a cold surface in one corner of the room and expect natural convection to eventually cool the room. However, this is likely to be very slow. Therefore blowers are used to force air in the room to make contact with the cold surfaces provided by the air conditioner, and get cooled in the process. This cold air is forced to mix with the rest of the air in the room and therefore helps cool the room quickly.

Radiation: This form of heat transfer involves the use of electromagnetic waves. Consider the transfer of heat from the sun to the earth. There is no direct physical contact between the two, therefore conduction is not possible. Except for the relatively thin layer of atmosphere around the earth, there is no fluid in the space between the sun and the earth, therefore convection is not possible. Energy carried through electromagnetic waves is the only way in which heat arrives from the sun to the earth.

Incidentally, once the heat arrives at the earth, the atmosphere around the earth uses convection to distribute the heat throughout the planet. The atmosphere around the earth ensures that the temperature difference between the side lit by the sun and the side that is facing away from the sun, is not significantly different. On the other hand, since there is no atmosphere around space craft, there is usually a very significant temperature difference between the side facing the sun and the side facing away from the sun.

With this background on the different modes of heat transfer, let us now look at thermal conduction using the Drude model. While we carry out this analysis we must keep one additional detail in mind – in metals heat conduction occurs both through the free electrons in the sample as well as the lattice vibrations present in the solid, which are called phonons. The lattice vibrations pass on the energy from atom to atom. In metallic systems, due to the presence of a large number of free electrons, the electronic contribution to thermal conductivity is the dominant contribution. In thermal insulators, the lattice contribution is significant. Since we are restricting our discussion to metals at the moment, we will only look at the electronic contribution to thermal conductivity.
To summarize at this stage, there are different modes of heat transfer, of which we are limiting our discussion to conduction. Further, conduction can occur through the electrons in the sample as well as lattice vibrations, and we are limiting our discussion to the conduction through electrons alone. For solid, metallic samples, these are reasonable limits to work within.

Thermal conductivity is denoted by K.
If the heat energy transferred by conduction is Q, and this heat is transferred in time t, through a cross sectional area A, and in response to a temperature gradient $\frac{\Delta T}{\Delta x}$, the conductivity is given by:

$$K = \frac{-\frac{Q}{t} \times \frac{1}{A}}{\frac{\Delta T}{\Delta x}}$$

Conductivity is the heat transferred per unit time, per unit cross sectional area, per unit temperature gradient.

The heat transferred per unit time, per unit cross sectional area is the flux of heat J_q. Therefore in differential form, for a 1 dimensional case, we can write the above equation as:

$$K = \frac{-J_q}{\frac{dT}{dx}}$$

Rearranging, we get:

$$J_q = -K\frac{dT}{dx}$$

The above equation is referred to as the Fourier's law of heat conduction.
We are interested in determining K based on the assumptions of the Drude model.
In order to determine K, let us consider the 1 Dimensional solid shown in Figure 8.2, and the analysis that follows:

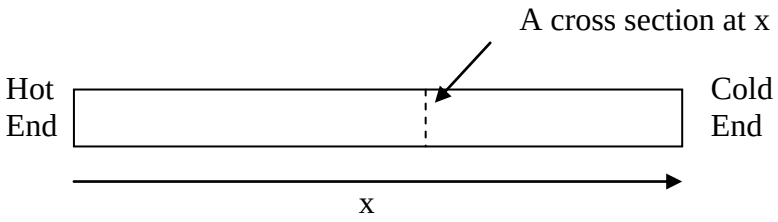


Figure 8.2: A schematic of a one dimensional sample, one end of which is hot and the other end of which is cold.

Let the left end of the sample be the hot end and the right end be the cold end as shown in Figure 8.2 above. In this case the energy E is a function of the temperature T, and the temperature T itself is a function of the position x. These can be denoted as follows:

$E[T]$ and $T(x)$

Which can be combined and written as

$$E[T(x)]$$

Consider a cross section at position x from the hot end of the sample. It is important to note that the atoms and electrons in the material do not 'know' which is the hot end and which is the cold end in the sample. Heat energy moves from the hot end of the sample to the cold end of the sample across this cross section as well as from the cold end to the hot

end – we perceive the net transfer of heat. In this context the process is similar to diffusion of chemical species.

We are therefore interested in determining the energy of the electrons that arrive at, and then cross, the position x. As indicated earlier, electrons exchange energy only through collisions with other electrons. Therefore, we need to determine the last collision that the electron had before it arrived at x. The energy corresponding to that position, will be the energy that the electron will possess as it arrives at x. We had also noted as part of the Drude assumptions that there is a mean time between collisions τ, which is independent of position and velocity of the electrons.

So where did the electrons have their last collision before they arrived at the position x? Since they are moving with the velocity v, and have a mean time between collisions τ, the electrons coming from the hot end had their last collision at $x - v\tau$, and the electrons coming from the cold end had their last collision at $x + v\tau$.

The energies corresponding to these two positions are:

$E[T(x - v\tau)]$ and $E[T(x + v\tau)]$ respectively

Since there are N_f free electrons per unit volume at all positions within the sample, and since the electrons are assumed to move randomly, at each position we can assume that half the electrons are moving towards the hot end of the sample and half the electrons are moving towards the cold end of the sample. In other words, we are not imposing any directional preference for the movement of the electrons.
Since the velocity of the electrons is v, the flux of heat from the hot end to the cold end, across the cross section at x, is given by:

$$J_{hot\ to\ cold} = \frac{N_f}{2} vE[T(x - v\tau)]$$

Dimensionally this is

$$m^{-3} \times \frac{m}{s} \times J = J/m^2/s$$

Which is consistent with the flux of heat.

Similarly, the flux of heat from the cold end to the hot end, across the cross section at x, is given by:

$$J_{cold\ to\ hot} = \frac{N_f}{2} vE[T(x + v\tau)]$$

Subtracting the two, we get the net transfer of heat from the hot end to the cold end of the sample, across the cross section at x.

$$J_{net} = J_{hot\ to\ cold} - J_{cold\ to\ hot}$$

Therefore,

$$J_{net} = \frac{N_f}{2} vE[T(x - v\tau)] - \frac{N_f}{2} vE[T(x + v\tau)]$$

For small changes in x, we can use the approximation

$$E[T(x-v\tau)] = E[T(x)] - v\tau\frac{dE}{dx}$$

Similarly,

$$E[T(x+v\tau)] = E[T(x)] + v\tau\frac{dE}{dx}$$

Substituting back in the expression for J_{net}, we get

$$J_{net} = \frac{N_f}{2}v\left\{E[T(x)] - v\tau\frac{dE}{dx} - \left(E[T(x)] + v\tau\frac{dE}{dx}\right)\right\}$$

Or

$$J_{net} = \frac{N_f}{2}v\left\{E[T(x)] - v\tau\frac{dE}{dx} - E[T(x)] - v\tau\frac{dE}{dx}\right\}$$

Therefore,

$$J_{net} = \frac{N_f}{2}v\left\{-2v\tau\frac{dE}{dx}\right\}$$

Which leads to

$$J_{net} = -N_f\, v^2\, \tau\frac{dE}{dx}$$

Since energy E is a function of the temperature T, and the temperature T is a function of the position x, we can rewrite the above as:

$$J_{net} = -N_f\, v^2\, \tau\frac{dE}{dT}\,\frac{dT}{dx}$$

Since E is the energy of a single electron, $\frac{dE}{dT}$ is the contribution to specific heat from the single electron, and since N_f is the number of free electrons per unit volume, $N_f\frac{dE}{dT}$ is the specific heat per unit volume associated with the electrons, and is therefore denoted using C_V^e.

Therefore:

$$J_{net} = -\,v^2\tau C_V^e\frac{dT}{dx}$$

Since the electrons are moving randomly, only one third of the electrons can be expected to travel along the x direction. Therefore, in order to generalize from the 1 dimensional case that we have considered, to a realistic 3 dimensional sample, we have to recognize that only one third of the heat flux will occur in the x direction. Further, we have derived the above using the velocity of a single electron. In reality, the large collection of electrons in a solid will have a distribution of velocities associated with them. Therefore, as we average over the collection of electrons, we have to use the mean of the square of velocities of the electrons, $\langle v^2\rangle$, rather than the square of the velocity of a single electron, v^2.

Taking into account the above two observations, we now rewrite the equation as follows:

$$J_{net} = -\frac{1}{3}\langle v^2\rangle\tau C_V^e\frac{dT}{dx}$$

Comparing this expression with Fourier's law of heat conduction:

$$J_q = -K\frac{dT}{dx}$$

We have the thermal conductivity of a metallic sample given by:

$$K = \frac{1}{3}\langle v^2 \rangle \tau C_V^e$$

The dimensions of K work out as follows:

$$\frac{m^2}{\mathrm{s}^2}\ \mathrm{s}\ \frac{\mathrm{J}}{\mathrm{m}^3\mathrm{K}}$$

Which works out to:

$$\mathrm{Js^{-1}m^{-1}K^{-1}}$$

Or:

$$\mathrm{Wm^{-1}K^{-1}}$$

In terms of the numerical value of the thermal conductivity, we can substitute the following into the expression for thermal conductivity:

$$\langle v^2 \rangle \approx 10^{10}\ \frac{m^2}{\mathrm{s}^2}$$

$$\tau \approx 10^{-14}\ s$$

$$C_V^e \approx 10^6\ \mathrm{J/(m^{-3}\ K)}$$

Therefore:

$$\mathrm{K} \approx 10^2\ \mathrm{Wm^{-1}K^{-1}}$$

The thermal conductivities of metallic conductors such as Ag, Cu, and Au, are in the same order of magnitude as the above prediction.